Big Data Analytics.

* **Matrix Multiplication using Map Reduce.**

$A_{ij} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B_{jk} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

→ **Mapper for Matrix A**

$A(K,V) = (i,k)(A, j, A_{ij})$ for all k.

Mapper for Matrix B

$B(K,V) = (i,k)(B, j, B_{jk})$ for all i.

* Tompute Mapper for matrix A :-

For K=1
$K=1;\ i=1,\ j=1 \Rightarrow (1,1)(A,1,A_{11}) \Rightarrow (1,1)(A,1,1)$
$K=1;\ i=1;\ j=2 \Rightarrow (1,1)(A,2,A_{12}) \Rightarrow (1,1)(A,2,2)$
$K=1;\ i=2;\ j=1 \Rightarrow (2,1)(A,1,A_{21}) \Rightarrow (2,1)(A,1,3)$
$K=1;\ i=2\ \ j=2 \Rightarrow (2,\cancel{1})(A,2,A_{22}) \Rightarrow (2,2)(A,2,4)$

For K=2
$K=2;\ i=1;\ j=1 \Rightarrow (1,2)(A,1,A_{11}) \Rightarrow (1,2)(A,1,1)$
$K=2;\ i=1;\ j=2 \Rightarrow (1,2)(A,2,A_{12}) \Rightarrow (1,2)(A,2,2)$
$K=2;\ i=2;\ j=1 \Rightarrow (2,2)(A,2,A_{21}) \Rightarrow (2,1)(A,\cancel{1},3)$
$K=2;\ i=2;\ j=2 \Rightarrow (2,2)(A,2,A_{22}) \Rightarrow (2,2)(A,2,4)$

* Tompute Mapper for Matrix B :-

For K=1
$K=1;\ i=1;\ j=1 \Rightarrow (1,1)(B,1,\cancel{A}_{11}) \Rightarrow (1,1)(B,1,5)$
$K=1;\ i=1;\ j=2 \Rightarrow (1,1)(B,2,\cancel{A}_{21}) \Rightarrow (1,1)(B,2,7)$
$K=1;\ i=2;\ j=1 \Rightarrow (2,1)(B,1,B_{11}) \Rightarrow (2,1)(B,1,5)$
$K=1;\ i=2;\ j=2 \Rightarrow (2,1)(B,2,B_{21}) \Rightarrow (2,1)(B,2,7)$

For K=2
$K=2;\ i=1;\ j=1 \Rightarrow (1,2)(B,1,B_{12}) \Rightarrow (1,2)(B,1\ 6)$
$K=2;\ i=1;\ j=2 \Rightarrow (1,2)(B,2,B_{22}) \Rightarrow (1,2)(B,2,8)$
$K=2;\ i=2;\ j=1 \Rightarrow (2,2)(B,1,B_{12}) \Rightarrow (2,2)(B,1,6)$
$K=2;\ i=2;\ j=2 \Rightarrow (2,2)(B,2,B_{22}) \Rightarrow (2,2)(B,2,8)$

* Computing the Reducer.

Common pairs :- (1,1), (1,2), (2,1), (2,2) — Multiply last values

(1,1), Alist = { (A,1,1), (A,2,2) }
Blist = { (B,1,5), (B,2,7) }

Now $A_{ij} \times B_{jk} = \{ (1\times5) + (2\times7)] = \underline{19}$

Similarly, calculate for {(1,2), (2,1), (2,2)}

(1,2) ; Alist = { (A,1,1), (A,2,2) }
Blist = { (B,1,6), (B,2,8) }

Now $A_{ij} \times B_{jk} = 1\times6 + 2\times8 = 6+16 = 22$

(2,1) ; Alist = { (A,1,3), (A,2,4) }
Blist = { (B,1,5), (B,2,7) }

$= 3\times5 + 4\times7 = 15+28 = \underline{43}.$

(2,2) Alist = { (A,1,3) (A,2,4) }
Blist = { (B,1,6) (B,2,8) }

$= 6\times3 + 4\times8 = 18 + 32 = \underline{50}$